

EEE103/CSE161

Computer Programming

Lecture 01: Algorithms and Flowchart

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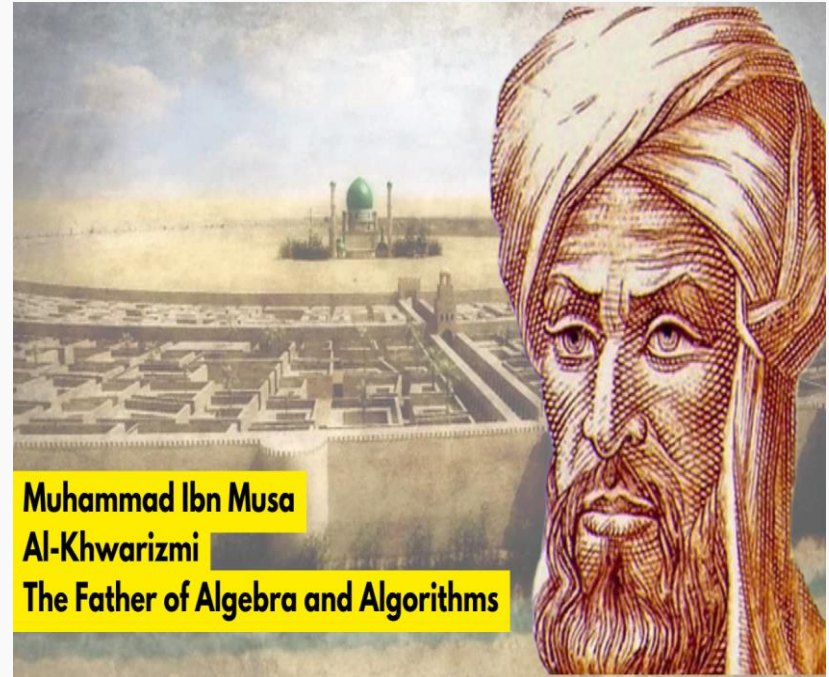
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What is an Algorithm?

- An algorithm is a set of step-by-step instructions that describe how to perform a task.
- It is made up of three basic building blocks:
✓ sequencing, selections and iteration.





What is a Flowchart?

A flowchart is a visual representation of an algorithm to solve a problem or to perform a particular task.



Steps in Problem Solving:

- **Develop a General Algorithm:**

Start by creating a broad algorithm using pseudocode/flowchart. Pseudocode is an informal and simplified language that resembles everyday English, making it easier for programmers to design and conceptualize algorithms.

- **Refine the Algorithm:**

Gradually refine the pseudocode into more detailed steps, bringing it closer to actual code in a specific programming language. This process involves breaking down each step into precise instructions that can be easily translated into a computer program.



What if we wanted to bake a cake?

- Firstly, we need to **sequence** the order of steps.
- Example
 1. Preheat the oven to 350°F.
 2. Mix flour, sugar, and eggs in a bowl.
 3. Pour the mixture into a baking pan.
 4. Bake for 30 minutes.



What if we wanted to bake a cake?

- **Selection** involves making decisions between different options.
- Example: Choosing Cake Type
 1. If you want a chocolate cake, add cocoa powder.
 2. If you want a vanilla cake, add vanilla extract.



What if we wanted to bake a cake?

- **Iteration** means repeating a step until a condition is met.
- Example: Checking Cake Doneness
 1. Stir the batter until smooth.
 2. Check if the cake is baked by inserting a toothpick. If it comes out clean, the cake is done. If not, bake for another 5 minutes and check again.



Pseudocode and Algorithm

Example question 1:

Write an algorithm to determine if a customer is eligible for a discount and calculate the final price. The discount eligibility is based on the total purchase amount. If the total amount is \$100 or more, the customer receives a 10% discount. The final price is calculated after applying the discount, if eligible. (Total purchase amount to be taken as input)



Pseudocode and Algorithm

Pseudocode:

- *Input Total amount.*
- *Check the value of total amount.*
- *If total amount is below 100.*
Final price = Total amount.
- *Else*
Final price = Total amount $\times$ 0.90
- *Print Final price*



Pseudocode and Algorithm

Detailed Algorithm For Discount Eligibility and Calculating Final Price:

Step 1: Start

Step 2: Input Total_amount

Step 3: Check the value of Total_amount

Step 4: If Total_amount < 100, then go to Step 5, else go to Step 6

Step 5:

- Set Final_price = Total_amount
- Go to Step 7

Step 6:

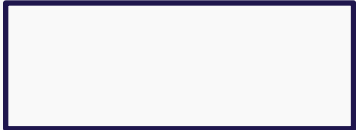
- Set Final_price = Total_amount $\times$ 0.90
- Go to Step 7

Step 7: Print Final_price

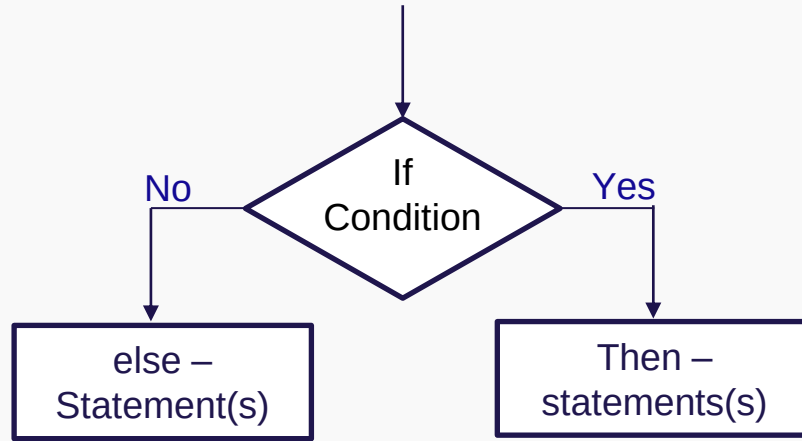
Step 8: End



Flowchart Symbols

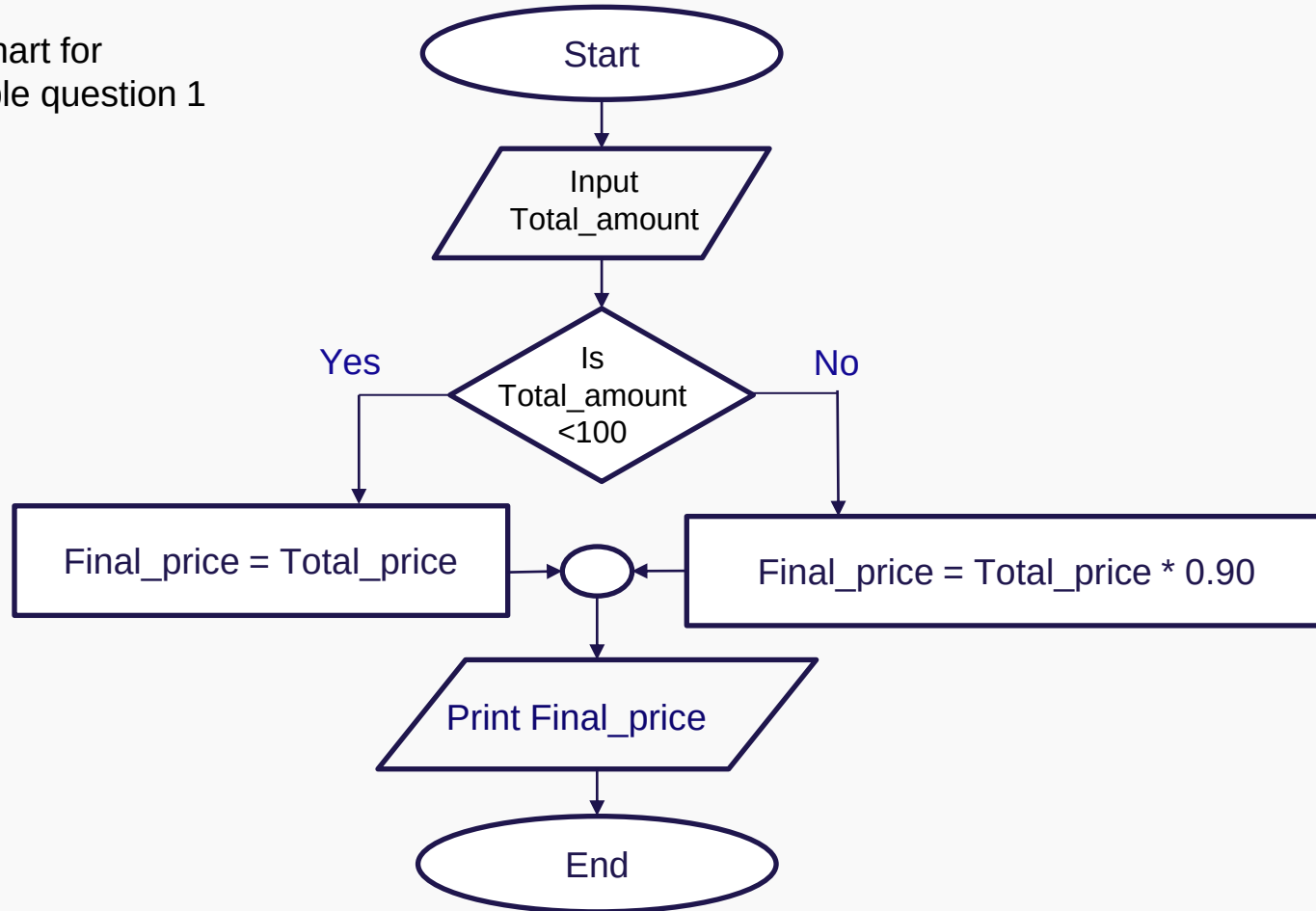
	Terminal symbol - indicates the beginning and end points of an algorithm. (more significant when we have a String of programs)
	Process symbol - shows an instruction other than input, output or selection. Shows calculations
	Input-output symbol - shows an input or an output operation.
	Selection/Decision symbol - shows a selection process for two-way selection. -> (for example if,else)
	Off-page connector - provides continuation of a logical path on another page.
	On-page connector - provides continuation of logical path at another point in the same page.

Flowchart-Selection Control Structure

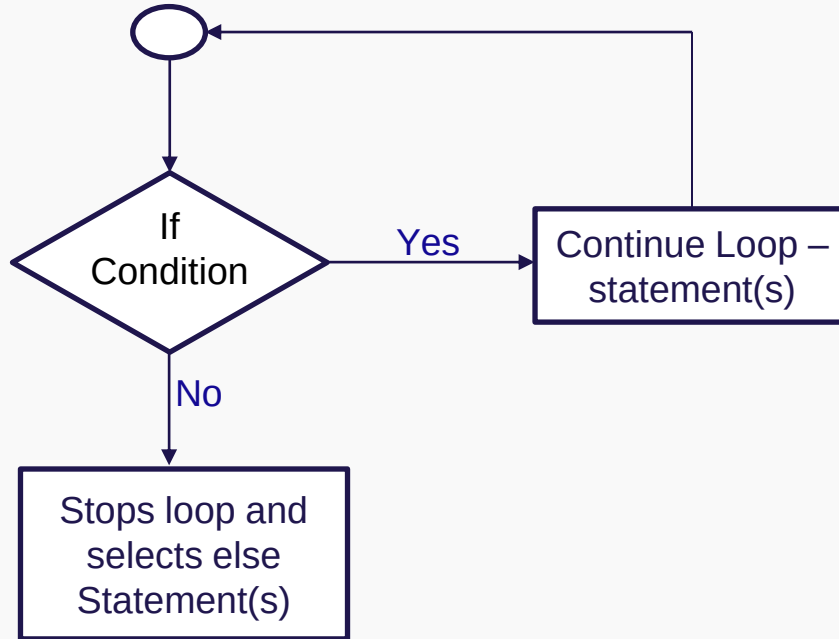




Flowchart for
example question 1



Flowchart-Repetition control structure





Example question 2:

Write an algorithm and draw a flowchart to calculate the sum of integers from 1 to 10.



Example question 2:

Write an algorithm and draw a flowchart to calculate the sum of integers from 1 to 10.

Algorithm:

Step 1: Start

Step 2: Count = 1, Sum = 1

Step 3: If Count $\leq$ 10, go to Step 4, else go to Step 6

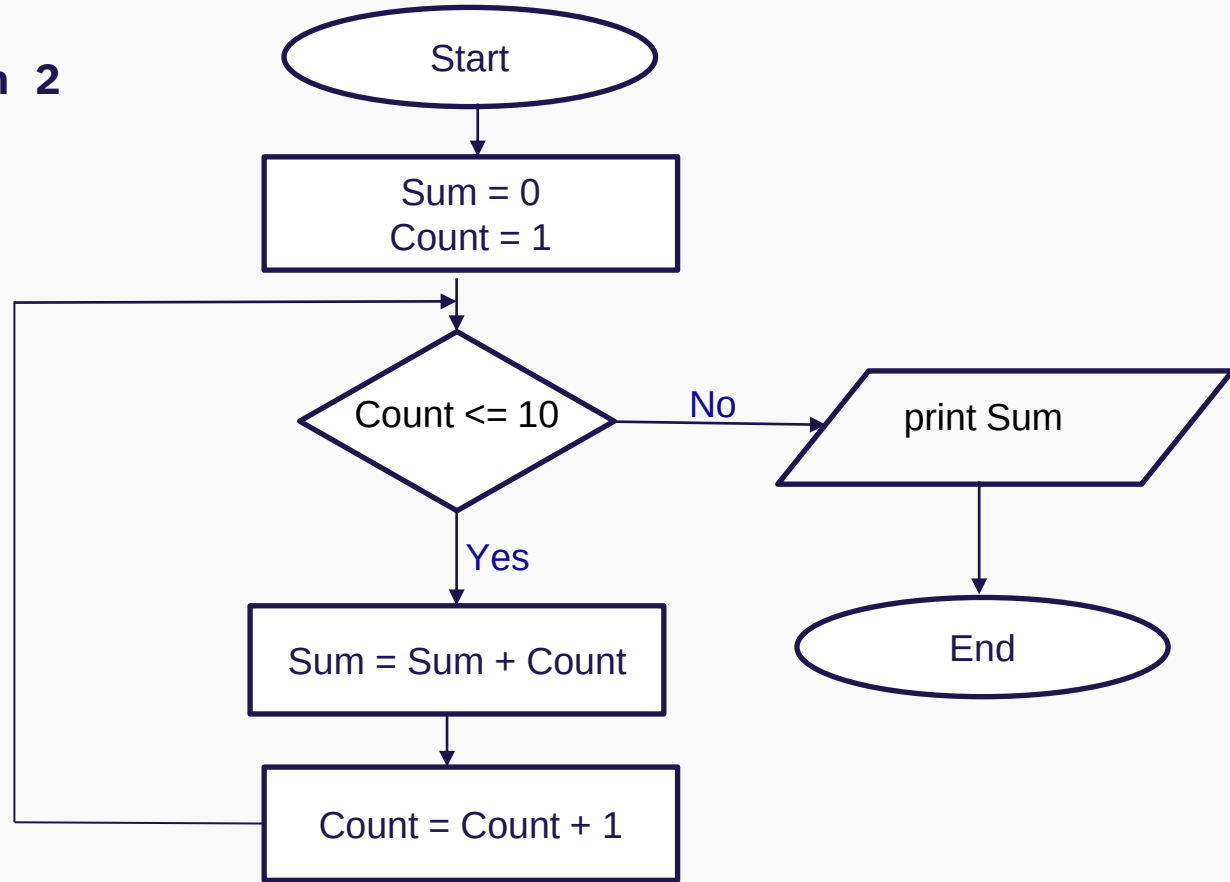
Step 4: Sum = Sum + Count, Count = Count + 1

Step 5: Return to Step 3

Step 6: Print Sum

Step 7: End

Flowchart for example question 2





Practice Questions:

1. Create an algorithm and a flowchart that will compute the sum of two numbers. If the sum is below or equal to twenty, two numbers will be entered again. If the sum is above 20, it will display the sum.
2. Create an algorithm and a flowchart to find the maximum of three input numbers.
3. Create an algorithm and a flowchart that will compute the sum of the following series up to n terms. (n will be provided as input). $1 + 4 + 7 + 10 + \dots$
4. Create an algorithm and a flowchart that will compute the sum of the following series $n*n$. (n will be provided as input).
5. $1*1 + 3*3 + 5*5 + 7*7 + \dots + n*n$



Thanks